



ROSHAN MOAZED

Brookline, MA 02445 | (617) 639-6340 | moazed.r@northeastern.edu

Education

Master of Science: Applied Mathematics

Northeastern University

Expected in May 2027

Boston, MA

Bachelor of Science: Applied Mathematics

University of Massachusetts - Boston

May 2023

Boston, MA

- Magna Cum Laude

Bachelor of Arts: Creative Writing

Brown University

May 2017

Providence, RI

- Honors

Skills

Languages & Tools: Python, SQL, Excel, VBA, Git, Jupyter, Streamlit

ML/DS: scikit-learn, XGBoost, TensorFlow, Forecasting, NLP, Recommendation Systems, Computer Vision

Libraries: pandas, NumPy, SciPy, Matplotlib, Plotly

Analytics: A/B Thinking, KPI reporting, Dashboards, Feature Engineering, Model Evaluation

Other: OpenAI API, Pinecone, Tableau, PowerBI, ArcGIS Pro

Experience

Data Analyst

Conservation Resources

Aug 2023 - Current

Exeter, NH

- Built SQL- and Python-driven reporting systems for monthly and quarterly portfolio performance reviews used by leadership and investors.
- Developed predictive models for agricultural portfolio performance, crop yield forecasting, and asset allocation optimization.

- Developed machine learning models (linear regression, random forest, logistic regression) to forecast apple orchard crop yields using historical weather and production data, achieving ~83% predictive accuracy.
- Designed CO2 emissions and sequestration models across 20+ managed properties for annual sustainability reporting.
- Connected SQL databases to vector search tools (Pinecone) and built an internal natural-language chatbot to improve non-technical data access.
- Produced dashboards, PowerPoint presentations, and executive summaries translating analytics into business decisions.
- Performed geospatial analysis in ArcGIS Pro across land assets and conservation properties.

Customer Satisfaction Analyst

Dec 2022 - Jun 2023

AI Lean

Boston, MA

- Built customer support analytics dashboards and reporting systems across multiple datasets.
- Reduced average ticket response time by 47% and resolution time by 54% through operational analytics and workflow improvements.
- Generated executive-level reporting on service KPIs, customer satisfaction, and internal team performance.
- Managed high-volume Zendesk data environments and SLA monitoring.

Projects

Retail Demand Forecasting & Inventory Optimization

- Built an end-to-end XGBoost demand forecasting platform using the Favorita Grocery Sales dataset across multiple product families.
- Engineered lag, rolling-window, calendar, and business-driver features to outperform naive forecasting baselines.
- Developed Streamlit dashboard with inventory simulation, volatility-based stock recommendations, and scenario planning tools.

Ecommerce Recommender System

- Built recommendation engine using collaborative filtering on ecommerce interaction data.
- Improved Hit Rate@10 by ~4.3x over baseline recommendation approaches.
- Deployed interactive product recommendation dashboard in Streamlit.

Brain Hemorrhage Segmentation & Classification

- Developed multi-view deep learning pipeline using U-Net style architectures for medical image segmentation.
- Performed advanced preprocessing, mask generation, training workflows, and deployment through interactive app (Streamlit).

Accomplishments

- 16 poetry publications to date